equicenta® CTM

Supplementary Materials

Companion appendix to the peer-reviewed clinical trial in Frontiers in Veterinary Science, 2026 — three supplementary figures and six supplementary tables.

"Horse First, Sport Second."

Equine Performance Labs' guiding principle for equicenta® CTM development.

PRIMARY PUBLICATION

Bertone AL, Davern A, Sutter WW, et al. Cryopreserved umbilical allograft (equicenta® CTM) improves clinical outcomes compared with clinician-selected standard care in performance horses: a pragmatic prospective controlled trial. Front. Vet. Sci. 2026. doi:10.3389/fvets.2026.1833933

Supplementary Figures

Three figures reproduced from the peer-reviewed publication: case-level imaging & score trajectories (S1), convergent treatment-effect signal (S2), and cumulative improvement in lameness, pain, and swelling (S3).

SUPPLEMENTARY FIGURES · S1 to S3

IN THIS SECTION

SUPPLEMENTARY FIGURE S1 · BERTONE ET AL. 2026

Case A vs Case B — imaging modalities and score trajectories

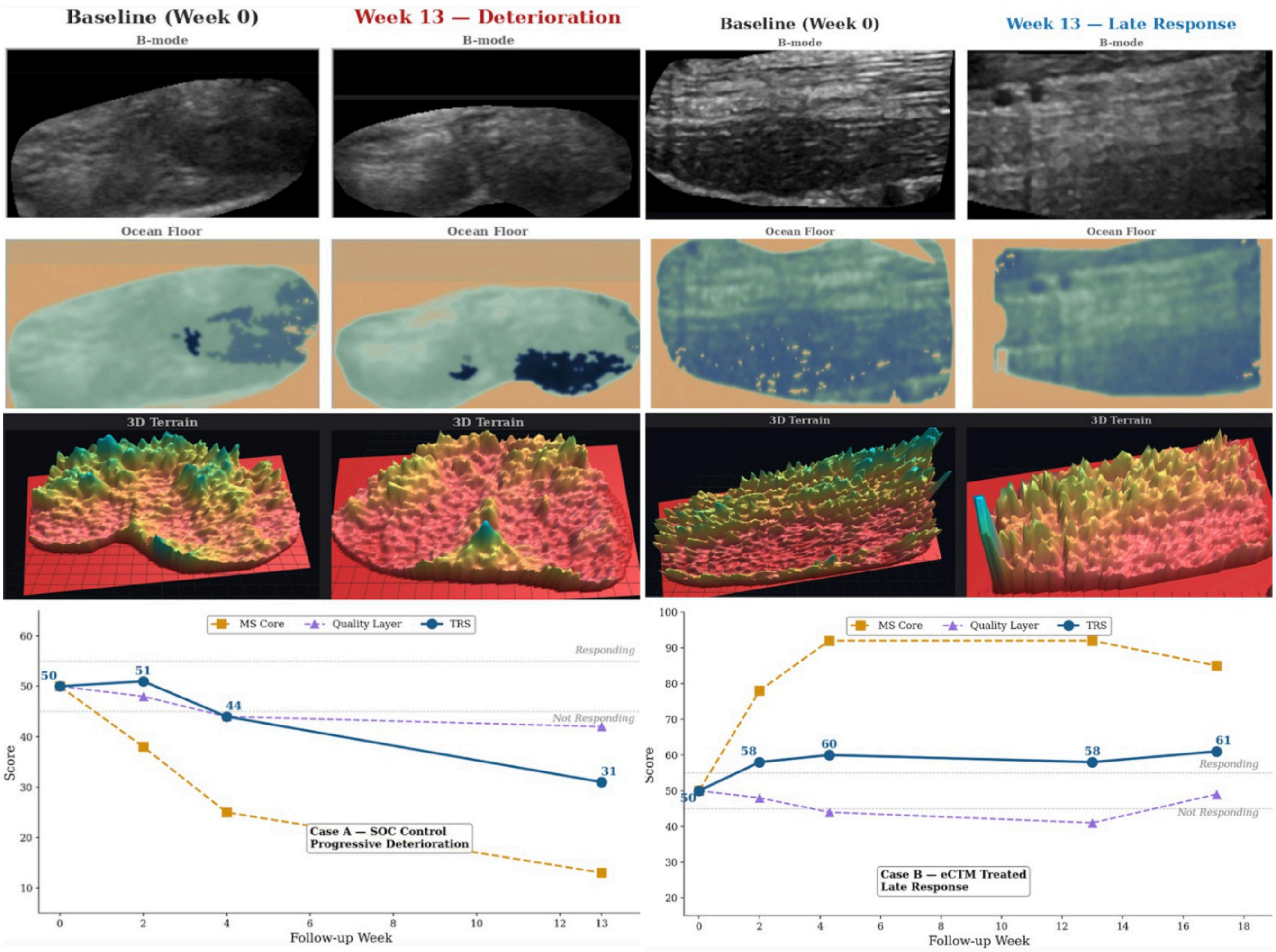


FIGURE CAPTION

Case A (SOC control) — a hind fetlock arthropathy case showing progressive deterioration in ultrasound B-mode, Ocean-Floor colorized mapping, and 3D terrain reconstruction from Baseline (Week 0) to Week 13, with corresponding declines in the tissue-response score (TRS), MS Core score, and Quality-Layer score. Case B (equicenta® CTM treated) — a proximal suspensory ligament case showing tissue substrate improvement on all three imaging modalities from Baseline (Week 0) through Week 13, with rising TRS and MS Core scores crossing the 'Responding' threshold and stabilizing at 26-week follow-up.

SUPPLEMENTARY FIGURE S2 · BERTONE ET AL. 2026

Convergent treatment-effect signal across modalities

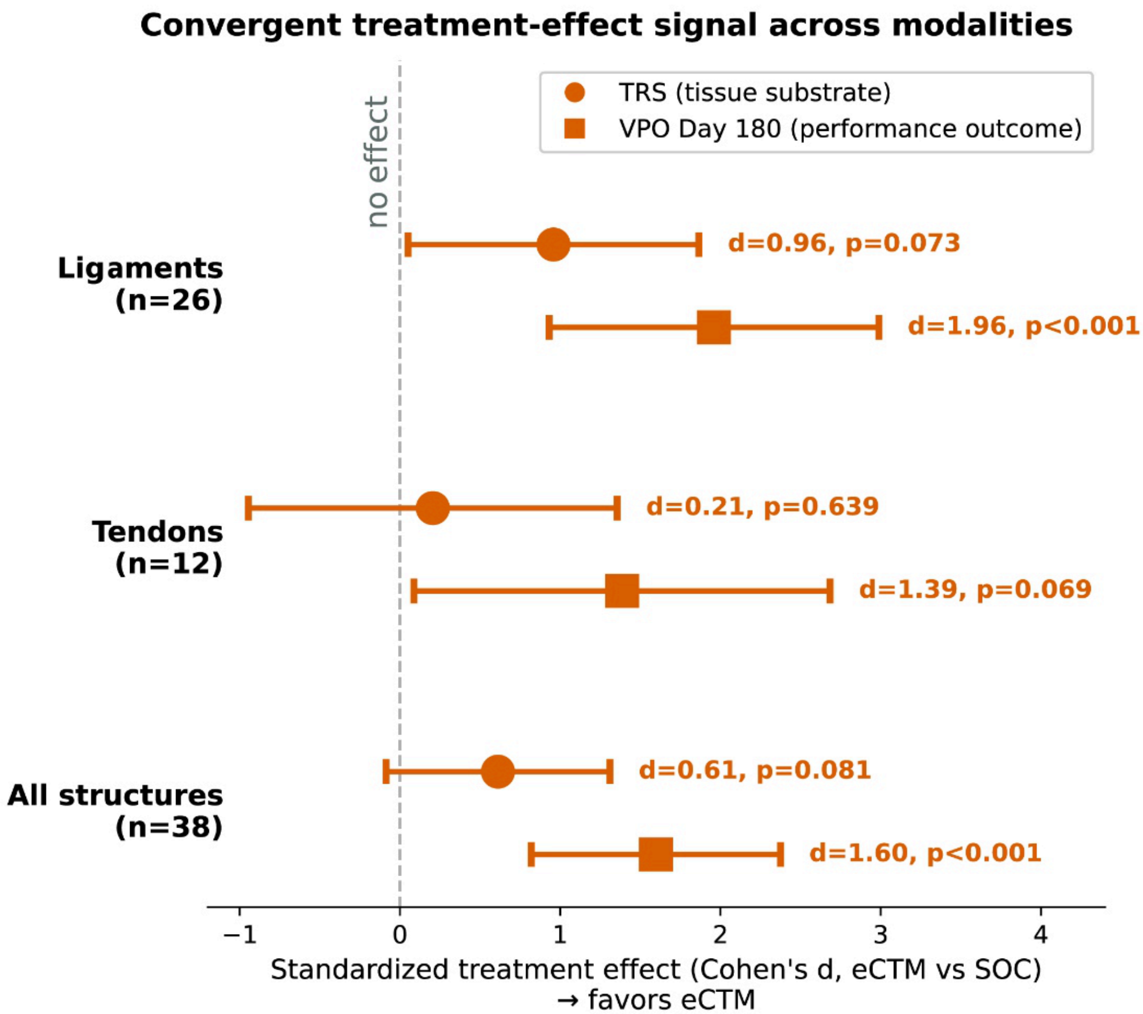
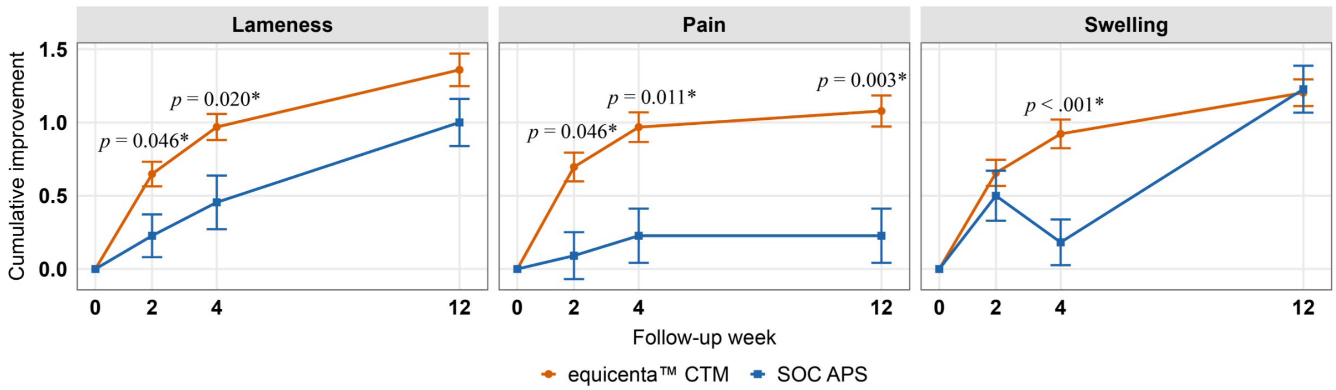


FIGURE CAPTION

Convergent treatment-effect signal across modalities. Standardized treatment effect (Cohen's d, equicenta® CTM vs SOC) for the tissue-substrate score (TRS, circles) and the Veterinary Performance Outcome at Day 180 (VPO, squares), plotted separately for Ligaments (n=26), Tendons (n=12), and All soft-tissue structures combined (n=38). Both modalities point in the direction favoring equicenta® CTM; VPO Day 180 effects are large and statistically significant in ligaments (d=1.96, p<0.001) and in all structures combined (d=1.60, p<0.001). TRS effect sizes are of similar magnitude but with wider confidence intervals given tissue-level imaging noise.

SUPPLEMENTARY FIGURE S3 · BERTONE ET AL. 2026

Cumulative improvement — lameness, pain, and swelling (0–12 weeks)



	Wk 0	Wk 2	Wk 4	Wk 12
equicenta™ CTM	128	125	128	128
SOC APS	22	22	22	22

	Wk 0	Wk 2	Wk 4	Wk 12
equicenta™ CTM	128	125	126	128
SOC APS	22	22	22	22

	Wk 0	Wk 2	Wk 4	Wk 12
equicenta™ CTM	128	125	128	128
SOC APS	22	22	22	22

FIGURE CAPTION

Cumulative improvement over 12 weeks for lameness, pain, and swelling in horses treated with equicenta® CTM (orange) versus standard-of-care APS (blue). Separation from SOC emerges at Week 2 for lameness (p=0.046) and pain (p=0.046), and at Week 4 for swelling (p<0.001). Improvement continues through Week 12 for lameness (p=0.020 at Wk 4) and pain (p=0.003 at Wk 12). Sample sizes: equicenta® CTM n=125–128 per visit; SOC APS n=22 per visit. Error bars are standard error of the mean. Asterisks denote statistically significant differences vs SOC at that visit.

Supplementary Tables

Six tables reproduced from the peer-reviewed publication: baseline weight & body-condition data, regional recruitment distribution, case-mix by arm, adverse-event details, and rescue-injection records.

SUPPLEMENTARY TABLES · S1 to S6

IN THIS SECTION

SUPPLEMENTARY TABLE S1 · BERTONE ET AL. 2026

Body weight of horses that completed the clinical trial (kg)

	SOC Control	equicenta® CTM	Overall
Week 0	496 ± 8.1	487 ± 5.0	492 ± 4.2
Week 12	513 ± 7.0	492 ± 4.4	503 ± 3.8
Mean by group	504.5	489.5	

SOC Control and equicenta® CTM did not statistically differ

SUPPLEMENTARY TABLE S2 · BERTONE ET AL. 2026

Mean body-condition score per horse (weeks 0–12)

	SOC Control		equicenta® CTM	
Body Score	*n*	%	*n*	%
3	1	2.1	0	0.0
4	7	14.9	7	7.7
5	23	48.9	61	67.0
6	10	21.3	18	19.8
7	6	12.8	4	4.4
8	0	0.0	1	1.1
Total	47		91	

SOC= standard of care; Mean body scores remained within the protocol range (3-8) for SOC and equicenta® CTM. Both equicenta®CTM and SOC had 5 as the most common body score.

SUPPLEMENTARY TABLE S3 · BERTONE ET AL. 2026

Distribution of practice by recruitment map (horses & conditions)

Region	Horses *n* (%)	Conditions *n* (%)	Joint *n* (%)	Ligament *n* (%)	Tendon *n* (%)
Kentucky	37 (26.8)	53 (27.2)	40 (75.5)	10 (18.9)	3 (5.7)
SOC	11 (29.7)	17 (32.1)	16 (94.1)	1 (5.9)	0 (0.0)
eCTM	26 (70.3)	36 (67.9)	24 (66.7)	9 (25.0)	3 (8.3)
North Carolina	35 (25.4)	44 (22.6)	23 (52.3)	15 (34.1)	6 (13.6)
SOC	18 (51.4)	21 (47.7)	11 (52.4)	8 (38.1)	2 (9.5)
eCTM	17 (48.6)	23 (52.3)	12 (52.2)	7 (30.4)	4 (17.4)
Ohio	14 (10.1)	21 (10.8)	14 (66.7)	2 (9.5)	5 (23.8)
SOC	6 (42.9)	9 (42.9)	6 (66.7)	1 (11.1)	2 (22.2)
eCTM	8 (57.1)	12 (57.1)	8 (66.7)	1 (8.3)	3 (25.0)
Texas	10 (7.2)	17 (8.7)	15 (88.2)	0 (0.0)	2 (11.8)
SOC	5 (50.0)	10 (58.8)	10 (100.0)	0 (0.0)	0 (0.0)
eCTM	5 (50.0)	7 (41.2)	5 (71.4)	0 (0.0)	2 (28.6)
Florida	42 (30.4)	60 (30.8)	12 (20.0)	36 (60.0)	12 (20.0)
SOC	7 (16.7)	10 (16.7)	3 (30.0)	5 (50.0)	2 (20.0)
eCTM	35 (83.3)	50 (83.3)	9 (18.0)	31 (62.0)	10 (20.0)
Total	**138 (100.0)**	**195 (100.0)**	**104 (53.3)**	**63 (32.3)**	**28 (14.4)**
SOC	47 (34.1)	67 (34.4)	46 (68.7)	15 (22.4)	6 (9.0)
eCTM	91 (65.9)	128 (65.6)	58 (45.3)	48 (37.5)	22 (17.2)

Ranked in order of most to least number of horses per Project Veterinarian at the practices in the clinical study.

SUPPLEMENTARY TABLE S4 · BERTONE ET AL. 2026

Distribution of primary conditions per horse (SOC vs eCTM)

	SOC (n = 47)		eCTM (n = 91)	
	n	%	n	%
A. Primary Conditions*				
1 condition (single site).	30	63.8%	58	63.7%
2 conditions in one lame limb.	1	2.1%	11	12.1%
1 condition in two lame limbs	14	29.8%	19	20.9%
3 conditions	2	4.2%	3	3.3%
Total horses with >1 condition	17	36.2%	33	36.3%
Total horses	47	100.0%	91	100.0%

*There was no significant difference between groups in distribution of primary conditions.

SUPPLEMENTARY TABLE S5 · BERTONE ET AL. 2026

Adverse events by horse and by report — all injections

	Severity	Report #	Horse #	Practice	Products	Breeds	Discipline	Diagnosis	Signs	Wk 12 Lameness	Wk 26 VPO
Veterinarian Scores	Serious / SOC	2	1	P2	PAAG	WB	Jumper	LH/RH: Bilateral hind fetlock arthritis	Lameness Swelling	3	euthanized
	Nonserious / SOC	4	1	P6	PAAG	TB	Hunter	RH: tarsocrural arthritis	Marked effusion Lameness	3	2
	Serious / eCTM	0	0	none	none	none	none	none	none	none	none
	Nonserious / eCTM	3	3	P1	eCTM	Hackey	Driving	LH: Stifle arthritis	Lame Swelling	0	4
				P3	eCTM	STDB	Racing	LF: SDF tendonitis	Lameness	0	3
				P8	eCTM	QH	Racing	RF: DDF tendonitis (insertion)		4	2
Total Veterinarian Scores	***SOC***	**6**	**2**								
	eCTM	**3***	**3**								
Central Review	Serious SOC	9	4†	P2	Fat MSC	QH	Western	LH: Chronic active PSD	Lame Swollen	1	3
				P6	ACS	WB	Hunter	RF: Coffin joint arthritis	Painful	0	4
				P6	PAAG	TB	Hunter	RH: tarsocrural arthritis		3	2
				P2	PAAG	WB	Jumper	LH/RH: Fetlock arthritis		3	euthanized
	Nonserious SOC	4	4	P2	APS	QH	Western	LH: Chronic active PSD	Lame	1	3
				P2	APS	WB	Eventing	RF: PSD desmopathy	Lame	2	2
				P7	PRP	WB	Hunter	LH/RH: Suspensory branch desmitis	Swelling	0	1
				P6	PAAG	TB	Hunter	RH: tarsocrural arthritis	Effusion	3	2
	Serious eCTM	1	1	P2	eCTM	QH	Western	RH: tarsocrural arthritis	Lame Effusion	3	2
	Nonserious eCTM	4	4†	P2	eCTM	TB	Eventing	RH: LSB desmopathy	Swelling	4	2
				P1	eCTM	Hackey	Driving	LH: Stifle arthritis	Lame	0	4
				P3	eCTM	STDB	Racing	LF: SDF tendonitis	Swelling	0	3
				P8	eCTM	QH	Racing	RF: DDF tendonitis (insertion)	Lameness	4	2
Total Central Review	***SOC***	**13**	**8**								
	eCTM	**5***	**5****								

*Reported Adverse Reactions by the veterinarian ($p<0.04$) and central review ($p<0.002$) were less frequent in eCTM than eSOC.
** After central review, fewer horses had an adverse reaction reported in eCTM than SOC ($p<0.04$)
† One horse was a rescue injection
PAAG=polyacrylamide gel, WB=Warmblood, TB=Thoroughbred, QH = Quarter Horse, STDB=Standardbred, MSC=mesenchymal stromal cell, ACS=autologous conditioned serum, PRP=platelet-rich plasma
DDF=deep flexor tendon; SDF=superficial flexor tendon; LSB=lateral suspensory branch; PSD=proximal suspensory desmitis

SUPPLEMENTARY TABLE S6 · BERTONE ET AL. 2026

Rescue injections at the target site (per protocol)

Rescue Injections	Lameness at Rescue	Limb	Diagnosis	Product
Rescue — SOC→SOC				
	3	RH	Tarsocrural Arthritis	ACS to PAAG
	2	LF	SDF Tendonitis.	APS to Fat MSC
Rescue —SOC → eCTM — crossover				
	4	RH	Hip Arthritis	PRP
	3	LH	Suspensory Desmitis	HA and PSGAG
	2	LH	Suspensory Desmitis	Fat MSC
	3	LH	Stifle Arthritis	PRP
	1	LH	Fetlock Arthritis	HA
	1	LH	Fetlock Arthritis	PAAG
	2	RF	LSB Desmitis	PSGAG
	2	LF	Suspensory Desmitis.	PRP
	3	LH	Stifle Arthritis	Cultured Bone marrow MSC
	1	LH	Stifle Arthritis	PPP
	1	RH	Stifle Arthritis	PPP
Rescue —eCTM → eCTM				
	3	LH	Collateral Ligament Desmitis	eCTM
	3	LH	Tarsocrural Synovitis	eCTM
	4	RH	Hip Arthritis	eCTM
	2	RF	Coffin Joint Arthritis	eCTM
	3	RH	LSB Desmitis	eCTM
	3	LF	Coffin Joint Arthritis (cartilage damage on MRI)	eCTM
	3	RH	Tarsocrural Arthritis	eCTM
	4	LF	Carpus Arthritis	eCTM
	2	RF	Carpus Arthritis	eCTM

LSB = Lateral suspensory branch; ACS = Autologous Conditioned Serum; PAAG = polyacrylamide gel; APS = Autologous protein solution; PRP = Platelet-rich plasma; HA = Hyaluronic acid; PSGAG= polysulfated glycosaminoglycan; PPP= platelet-poor plasma; MSC=mesenchymal stromal cells;
The odds of hindlimbs (H) having rescue injections compared to forelimbs was 4.6:1 ($p<0.02$); The odds of SOC having more hindlimbs rescued than eCTM was 2.6:1 but did not statistically differ ($p>0.2$).

COLOPHON · BERTONE ET AL. 2026

About this document

WHAT'S INSIDE

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Supplementary Tables

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• Table S2 — Mean body-condition score per horse (weeks 0–12).

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CITE AS

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SCOPE OF CLAIMS

equicenta® CTM has not been shown to cure disease, and absolute safety has not been established. Results are from a single prospective controlled clinical study in performance horses; individual outcomes may vary. This document consolidates figures and tables from the primary publication for reference use by Equine Performance Labs field team.

"Horse First, Sport Second."

Equine Performance Labs · guiding principle for equicenta® CTM development.